

Part II - Prosper Loan Dataset

by S Sarveshwar

Investigation Overview

We will be investigating on the factors which can be used to predict default on credit and factors that influence Prosper rating.

Dataset Overview and Executive Summary

The dataset has 81 attributes across 113,937 loans, this investigation strictly utilizes a targeted subset.

```
# import all packages and set plots to be embedded inline
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# suppress warnings from final output
import warnings
warnings.simplefilter("ignore")
%matplotlib inline
sns.set_style('whitegrid')

df = pd.read_csv('prosperLoanData.csv')
```

Wrangling

```
loan_df = df[['BorrowerAPR',
               'ProsperScore',
               'LoanOriginalAmount',
               'EmploymentStatus',
               'StatedMonthlyIncome',
               'CreditScoreRangeLower',
               'CreditScoreRangeUpper',
               'IncomeRange',
               'LoanStatus',
               'Term']]

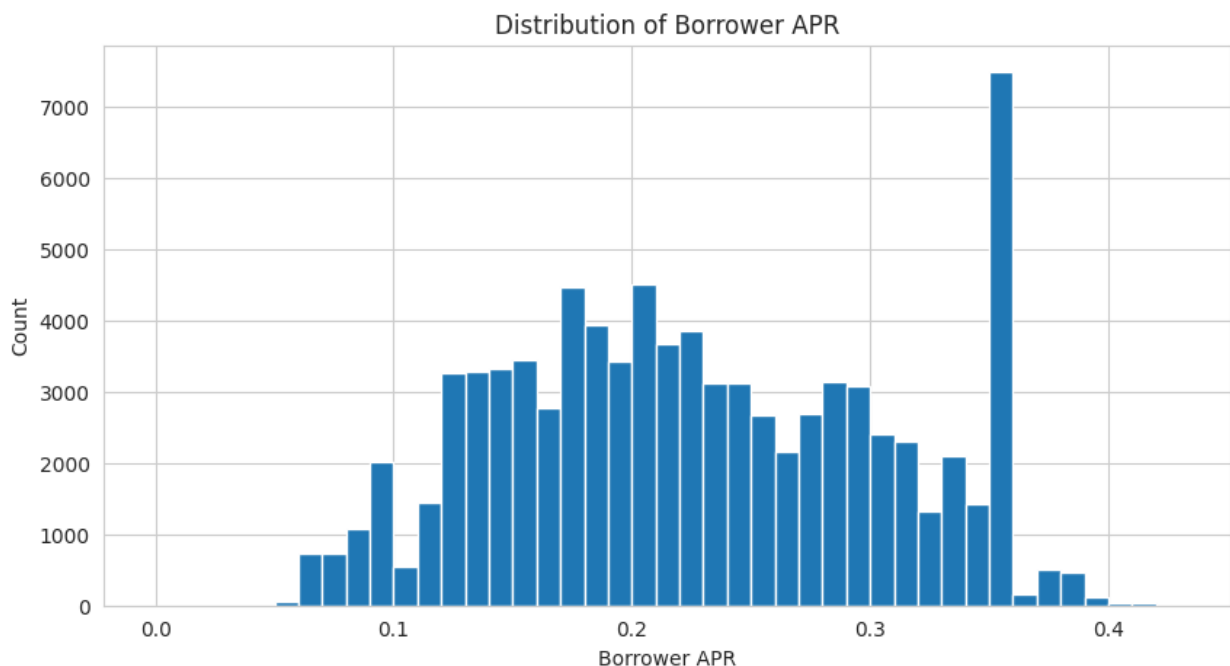
loan_df = loan_df.dropna()
loan_df['AverageCreditScore'] = (
    loan_df['CreditScoreRangeLower'] +
    loan_df['CreditScoreRangeUpper']
) / 2
```

Distribution of Borrower APR

APR ranges between 0% and 40%.

Most borrowers have APR between 10% and 25%.

```
plt.figure(figsize=[10,5])  
bins = np.arange(0, loan_df['BorrowerAPR'].max()+0.01, 0.01)  
plt.hist(data=loan_df, x='BorrowerAPR', bins=bins)  
plt.xlabel('Borrower APR')  
plt.ylabel('Count')  
plt.title('Distribution of Borrower APR');
```



Distribution of Loan Amount

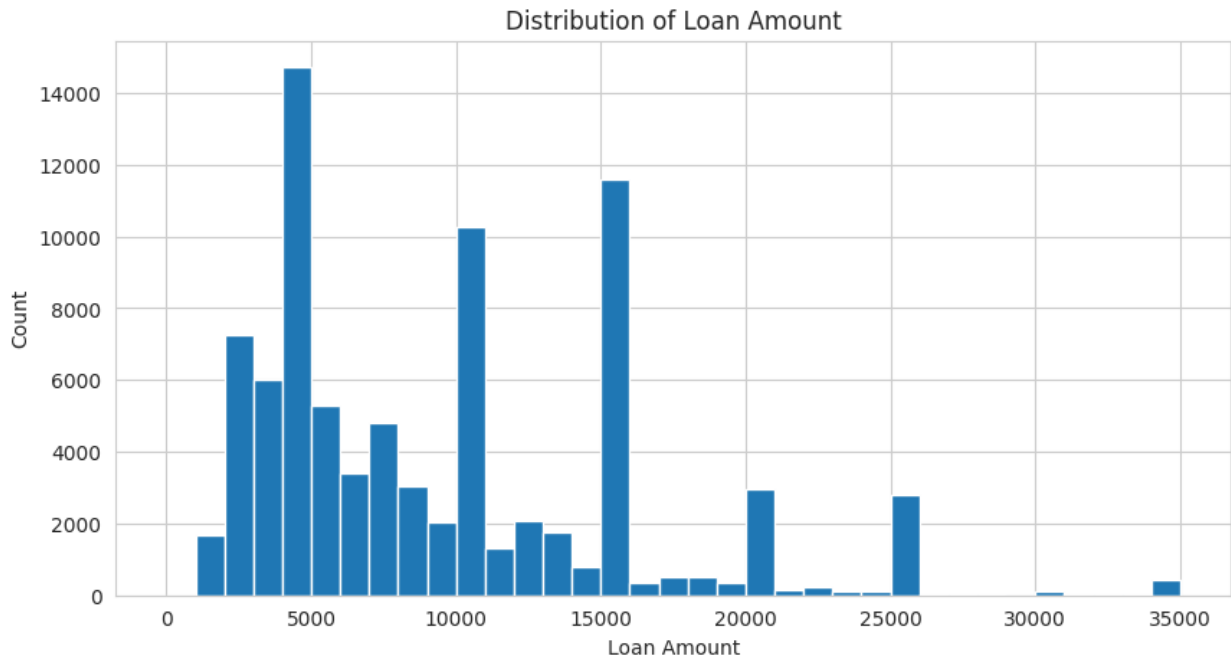
Most loans are below 15,000.

Distribution is right-skewed.

The mode is around 5000.

```
plt.figure(figsize=[10,5])  
bins = np.arange(0, loan_df['LoanOriginalAmount'].max()+1000, 1000)  
plt.hist(data=loan_df, x='LoanOriginalAmount', bins=bins)  
plt.xlabel('Loan Amount')
```

```
plt.ylabel('Count')
plt.title('Distribution of Loan Amount')
Text(0.5, 1.0, 'Distribution of Loan Amount')
```



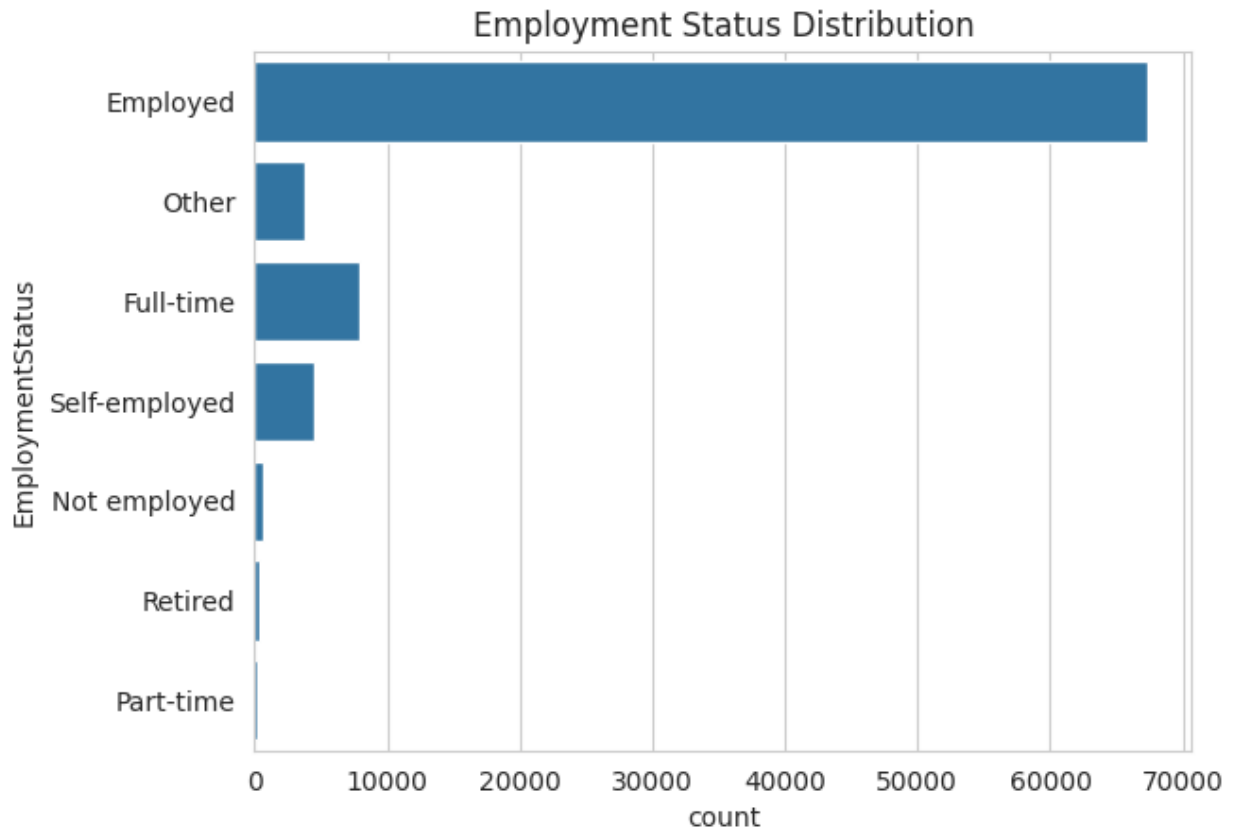
Employment Status

Most borrowers are employed

```
base_color = sns.color_palette()[0]

sns.countplot(data=loan_df,
              y='EmploymentStatus',
              color=base_color)

plt.title('Employment Status Distribution');
```

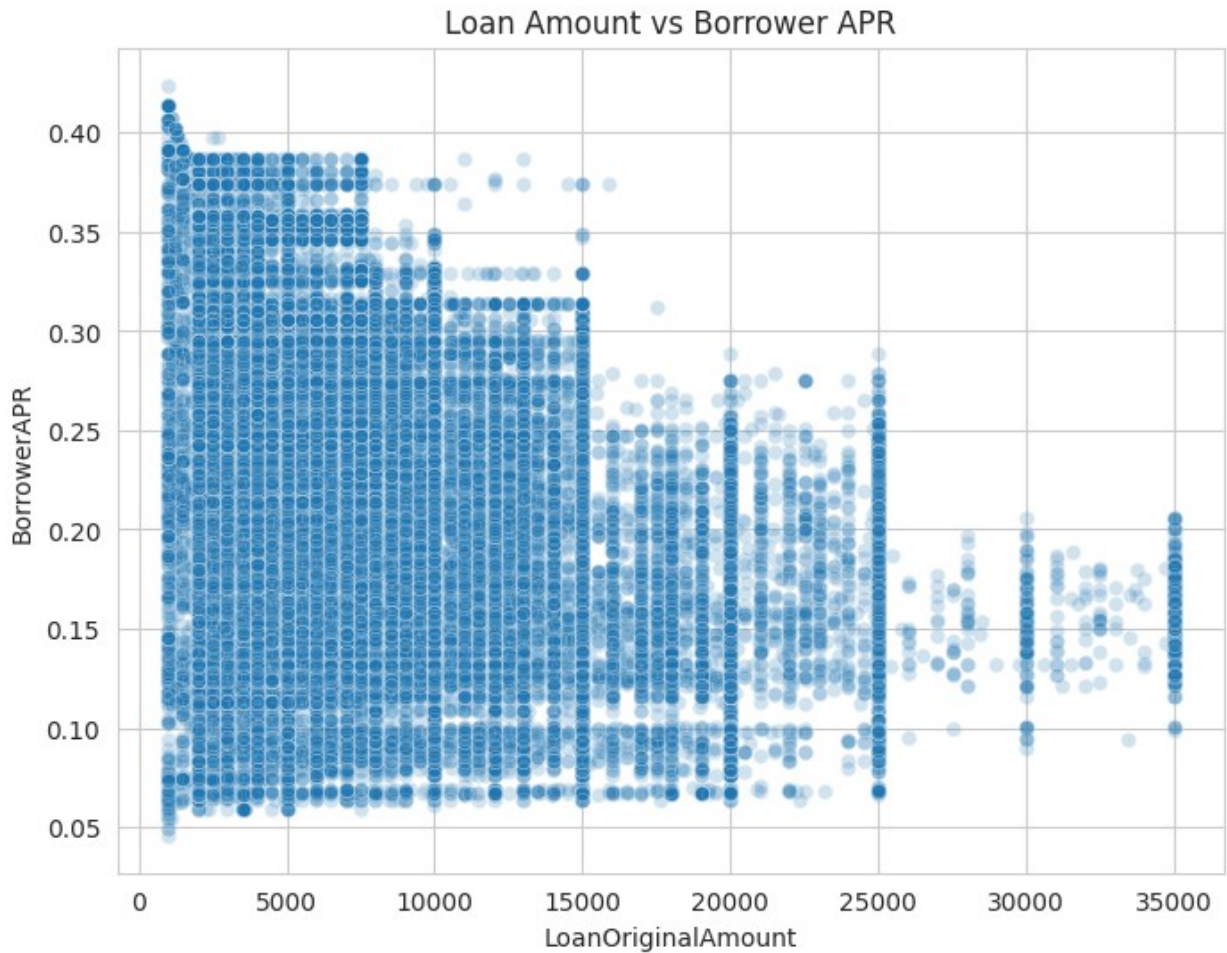


Loan Amount vs Borrower APR

Negative correlation observed.

Higher loan amounts generally have lower APR.

```
plt.figure(figsize=[8,6])  
  
sns.scatterplot(data=loan_df,  
                x='LoanOriginalAmount',  
                y='BorrowerAPR',  
                alpha=0.2)  
  
plt.title('Loan Amount vs Borrower APR')  
Text(0.5, 1.0, 'Loan Amount vs Borrower APR')
```

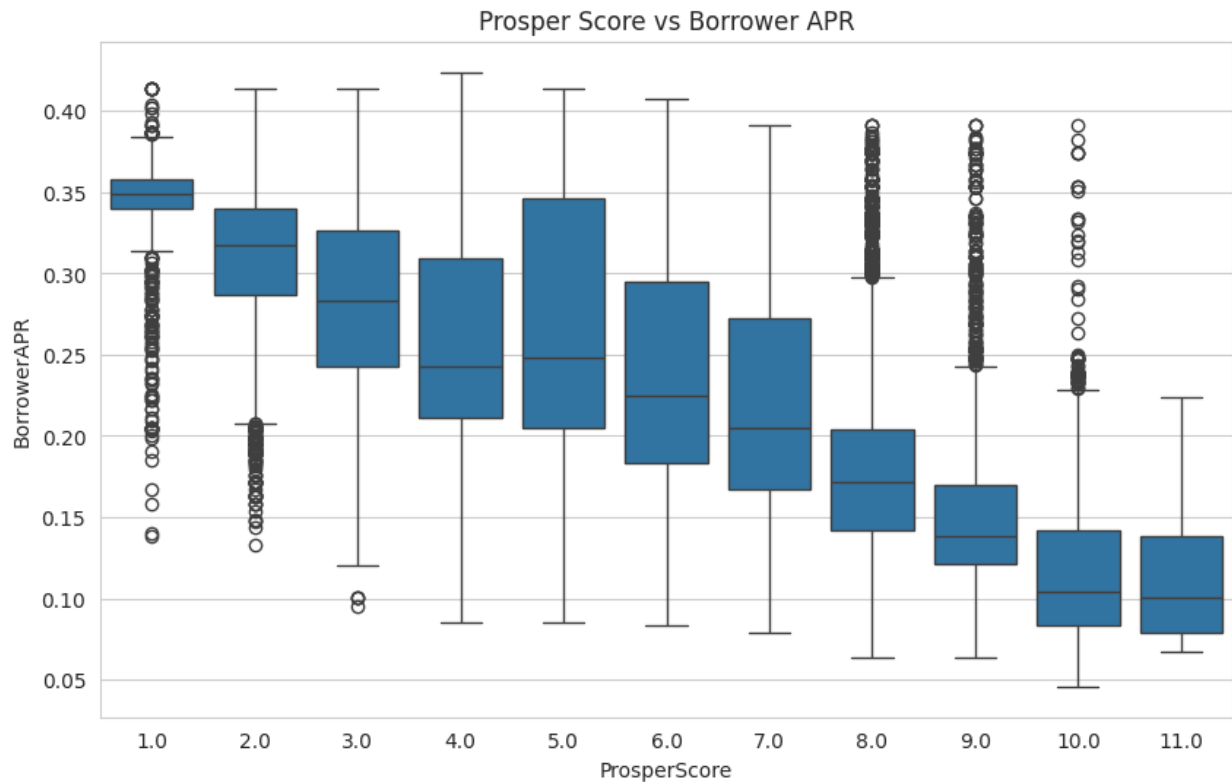


Prosper Score vs Borrower APR

Higher Prosper Score leads to lower APR.

Riskier borrowers receive higher interest rates

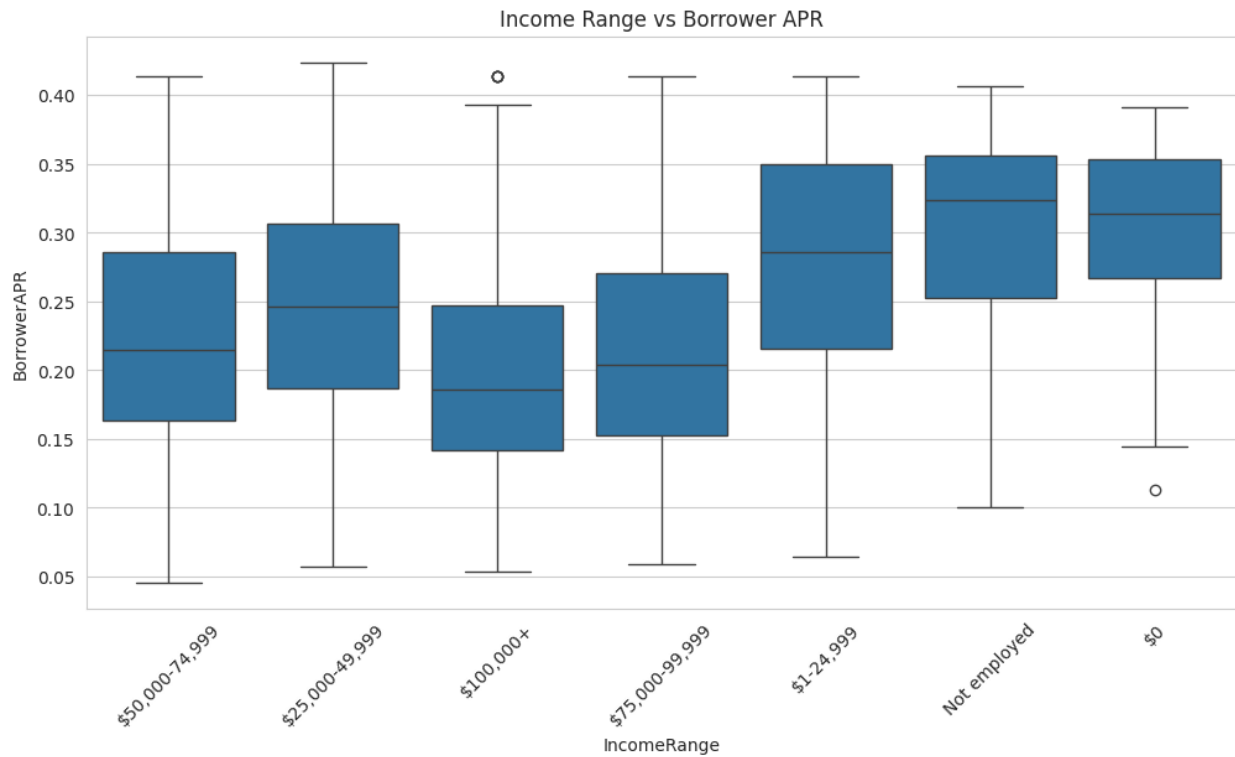
```
plt.figure(figsize=[10,6])  
sns.boxplot(data=loan_df,  
            x='ProsperScore',  
            y='BorrowerAPR')  
  
plt.title('Prosper Score vs Borrower APR')  
Text(0.5, 1.0, 'Prosper Score vs Borrower APR')
```



Income Range vs Borrower APR

Higher-income borrowers tend to receive lower APR

```
plt.figure(figsize=[12,6])  
sns.boxplot(data=loan_df,  
            x='IncomeRange',  
            y='BorrowerAPR')  
  
plt.xticks(rotation=45)  
plt.title('Income Range vs Borrower APR')  
Text(0.5, 1.0, 'Income Range vs Borrower APR')
```



APR vs Loan Amount by Prosper Score

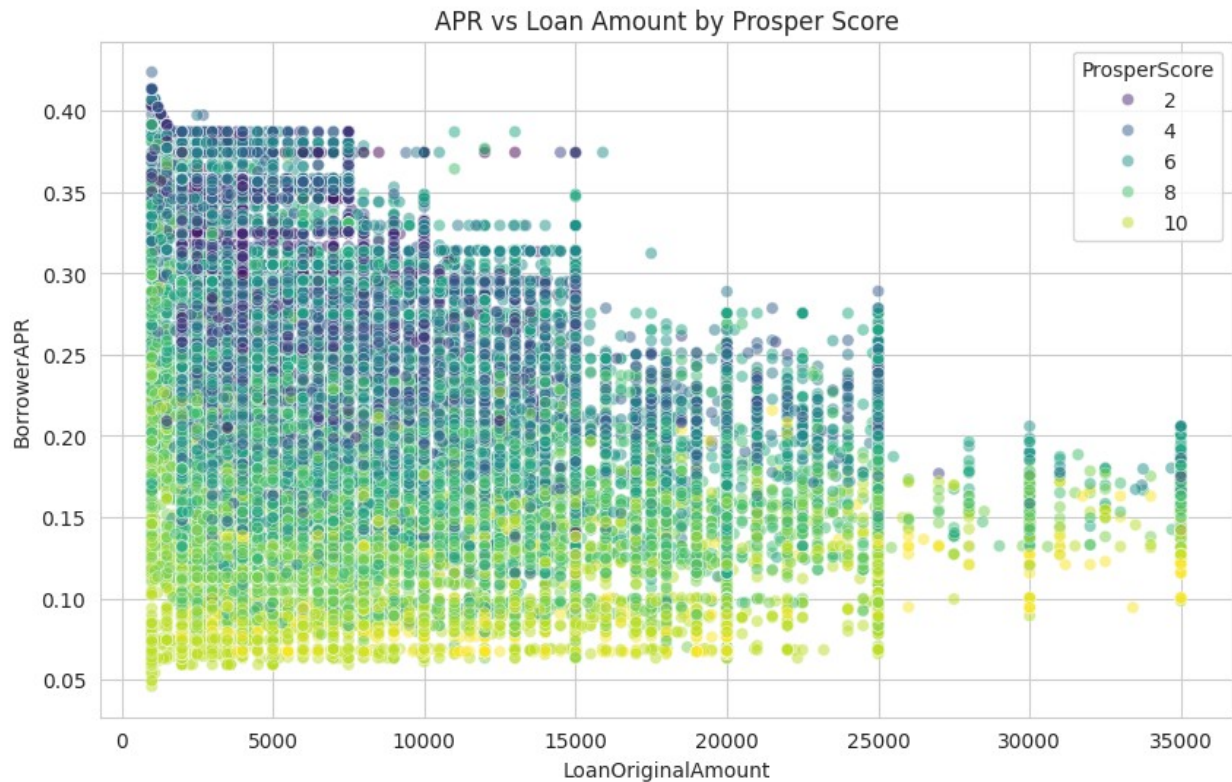
Borrowers with high Prosper Scores obtain lower APR even for large loans.

Low-score borrowers experience high APR.

```
plt.figure(figsize=[10,6])

sns.scatterplot(data=loan_df,
                x='LoanOriginalAmount',
                y='BorrowerAPR',
                hue='ProsperScore',
                palette='viridis',
                alpha=0.5)

plt.title('APR vs Loan Amount by Prosper Score');
```



Monthly Income vs APR by Prosper Score

Income affects APR differently across risk categories.

Prosper Score strongly impacts loan pricing.

```
g = sns.FacetGrid(data=loan_df,
                  col='ProsperScore',
                  col_wrap=3)

g.map(plt.scatter,
      'StatedMonthlyIncome',
      'BorrowerAPR',
      alpha=0.2)

g.fig.suptitle('Income vs APR by Prosper Score',
              y=1.02)

Text(0.5, 1.02, 'Income vs APR by Prosper Score')
```


Income vs APR by Prosper Score

